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Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/908,498	Applicant(s) Fang et al.	
	Examiner HAMZA N ALGIBHAH	Art Unit 2441	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 10/07/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-15 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-15 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 10/07/2024 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Detailed Action

1. ***Claims 1-15*** are pending.

2. ***Claims 1-15 are rejected.***

Double Patenting

The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the "right to exclude" granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

Claims **1-15** are rejected on the ground of nonstatutory double patenting as being unpatentable over claims **1-10** of U.S. Patent No. 11,252,039. Although the claims at issue are not identical, they are not patentably distinct from each other because the instant claims have broader scope than the patent claims. In addition, the instant claims require the policy to be communicated to the CND from a second external device. UNAGAMI discloses wherein the policy is communicated to the CND from a second external device (UNAGAMI, paragraph 0052, wherein " In addition, the in-vehicle network system may further include an external communication electronic controller, and the updated rule information may be received by the gateway device in response to a receipt of the updated rule information by the external communication electronic controller from an external device, the updated rule information being transmitted on the first bus by the external communication electronic controller. Thus, rule information for detecting a malicious frame can be appropriately updated even if a device in an in-vehicle network system that communicates with an external device and a fraud-detection ECU have separate configurations").

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify BARRETT using UNAGAMI so that the policy is revived from an external device as claimed because this would have provided a way to update the in-vehicle gateway/firewall with new rules that can help handle new security issues which improves the security of the system.

Instant claims	Patent claims
1. A system, comprising:	1. A system, comprising:
- a vehicle having **at least one network zone**;	- a vehicle having **a first network zone and a second network zone of a different type than the first network zone**;
- a converged network device (CND) interposed between **the at least one network zone and at least one of an external communication portal or a first external device**,	- a converged network device (CND) interposed between t**he first network zone and the second network zone**,
the CND comprising:	the CND comprising:
- a policy manager circuit structured to interpret a policy comprising a network regulation description, **wherein the policy is communicated to the CND from a second external device**; and	- a policy manager circuit structured to interpret a policy comprising a network regulation description; and
- a configuration circuit structured to configure **a gatekeeper interface circuit in response to the policy**; and	- a configuration circuit structured to configure **a first network interface circuit in response to the network regulation description**, and to configure a gatekeeper interface circuit in response to the network regulation description, **wherein the first network interface circuit is structured to regulate communications between end**

	points of the first network zone and end points of the second network zone, and
wherein the gatekeeper interface circuit is structured to regulate communications between end points of **the at least one network zone** and the at least one of the external communication portal or the first external device.	wherein the gatekeeper interface circuit is structured to regulate communications between end points of **the first network zone with** at least one of an external communications portal or an external device.

Claims **1-15** are rejected on the ground of nonstatutory double patenting as being unpatentable over claims **1-8** of US Patent No. 11,349,717. Although the claims at issue are not identical, they are not patentably distinct from each other because the instant claims have a broader scope than corresponding to the patent system claims and it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to omit the additional limitation in the patent claims to achieve the instant claimed limitation.

Instant claims	Patent claims
1. A system, comprising:	1. A system comprising:
- a vehicle having **at least one network zone**;	- a vehicle having **a first network zone and a second network zone of a different type than the first network zone**; and
- a converged network device (CND) interposed between **the at least one network zone and at least one of an external communication portal or a first**	- a converged network device (CND) interposed between **the first network zone and the second network zone**,

external device, the CND comprising: - a policy manager circuit structured to interpret a policy comprising a network regulation description, **wherein the policy is communicated to the CND from a second external device**; and - a configuration circuit structured to configure **a gatekeeper interface circuit in response to the policy**; and wherein **the gatekeeper interface circuit** is structured to regulate communications between end points of **the at least one network zone and the at least one of the external communication portal or the first external device**.	the CND comprising: - a policy manager circuit structured to interpret a policy comprising a network regulation description; - a configuration circuit structured to configure at least **one network interface circuit** in response to the policy; and the at least one network interface circuit; wherein: **the at least one network interface circuit** is structured to regulate communications between end points **of the first network zone and end points of the second network zone**; **the policy comprises a permission value corresponding to at least one end point of the first network zone or the second network zone; and the permission value comprises at least one permission value selected from at least one of: a data collection permission value; a service publication permission value; a service subscription permission value; or an external communication permission value.**

	8. The system of claim 1, wherein the policy is communicated to the CND from an external device.

Claim Interpretation

The following is a quotation of 35 U.S.C. 112(f):

> (f) Element in Claim for a Combination. – An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The following is a quotation of pre-AIA 35 U.S.C. 112, sixth paragraph:

> An element in a claim for a combination may be expressed as a means or step for performing a specified function without the recital of structure, material, or acts in support thereof, and such claim shall be construed to cover the corresponding structure, material, or acts described in the specification and equivalents thereof.

The claims in this application are given their broadest reasonable interpretation using the plain meaning of the claim language in light of the specification as it would be understood by one of ordinary skill in the art. The broadest reasonable interpretation of a claim element (also commonly referred to as a claim limitation) is limited by the description in the specification when 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is invoked.

As explained in MPEP § 2181, subsection I, claim limitations that meet the following three-prong test will be interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph:

(A) the claim limitation uses the term “means” or “step” or a term used as a substitute for “means” that is a generic placeholder (also called a nonce term or a non-structural term having no specific structural meaning) for performing the claimed function;

(B) the term "means" or "step" or the generic placeholder is modified by functional language, typically, but not always linked by the transition word "for" (e.g., "means for") or another linking word or phrase, such as "configured to" or "so that"; and

(C) the term "means" or "step" or the generic placeholder is not modified by sufficient structure, material, or acts for performing the claimed function.

Use of the word "means" (or "step") in a claim with functional language creates a rebuttable presumption that the claim limitation is to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites sufficient structure, material, or acts to entirely perform the recited function.

Absence of the word "means" (or "step") in a claim creates a rebuttable presumption that the claim limitation is not to be treated in accordance with 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph. The presumption that the claim limitation is not interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, is rebutted when the claim limitation recites function without reciting sufficient structure, material or acts to entirely perform the recited function.

Claim limitations in this application that use the word "means" (or "step") are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action. Conversely, claim limitations in this application that do not use the word "means" (or "step") are not being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, except as otherwise indicated in an Office action.

This application includes one or more claim limitations that do not use the word "means," but are nonetheless being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, because the claim limitation(s) uses a generic placeholder that is coupled with functional language without reciting sufficient structure to perform the recited function and the generic placeholder is not

preceded by a structural modifier. Such claim limitation(s) is/are: "a policy manager circuit structured to interpret, a configuration circuit structured to configure, and a gatekeeper interface circuit structured to regulate" in claim 1. The corresponding structures and algorithms used to link to the above limitations are the structures and algorithms stated in the specification (for example paragraph 0101, 0152, 0291-0296, 0307-0310)

Because this/these claim limitation(s) is/are being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, it/they is/are being interpreted to cover the corresponding structure described in the specification as performing the claimed function, and equivalents thereof.

If applicant does not intend to have this/these limitation(s) interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph, applicant may: (1) amend the claim limitation(s) to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph (e.g., by reciting sufficient structure to perform the claimed function); or (2) present a sufficient showing that the claim limitation(s) recite(s) sufficient structure to perform the claimed function so as to avoid it/them being interpreted under 35 U.S.C. 112(f) or pre-AIA 35 U.S.C. 112, sixth paragraph.

Claim Rejections - 35 USC § 103

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

3. ***Claims 1-2 and 8-15 are rejected*** under 35 U.S.C. 103 as being anticipated by BARRETT et al (Pub. No.: US 2019/0394089 A1) in view of UNAGAMI et al (Pub. No.: US 2017/0026386 A1).

As per claim 1, BARRETT discloses a system, comprising:

- ***a vehicle having at least one network zone*** (BARRETT, Fig 1, 2, 3, paragraph 0021, 0027-0028 vehicle network, the internal network 100 can be a heterogeneous network. For example, a vehicle can integrate over 100 ECUs connected over multiple networks such as CAN (Control Area Network), LIN (Local Interconnect Network), FlexRay)***;***

- ***a converged network device (CND) interposed between the first network zone and at least one of an external communication portal or a first external device*** (BARRETT, Fig 2, 3, paragraph 0037, wherein for example the gateway 240 can be the converged network device which is between internal network zones and external network)***,***

the CND comprising:

- ***a policy manager circuit structured to interpret a policy comprising a network regulation description*** (BARRETT, Fig 2, paragraph 0037, inter-domain firewalling can ensure that only certain types of traffic can pass from the external network 205 into the trusted network 220 internal to the vehicle 202. An internal firewall, FW.sub.int, 230 is placed between the switch or router 210 of the the trusted network 220 and the gateway 240 of the IZ 225. An external firewall, FW.sub.ext, 250 is placed between the gateway 240 of the IZ 225 and the telecoms control unit 260)***;***

- ***a configuration circuit structured to configure a gatekeeper interface circuit in response to the policy*** {BARRETT, paragraph 0034, 0037, 0046, 0064 The firewall system of the vehicle network is configured based on the connectivity information. In some implementations, configuring a firewall system of the vehicle network includes configuring one or more firewalls in the vehicle network based on the connectivity information (e.g., the connectivity information recorded in Table 1 or 2). In some implementations, configuring one or more firewalls in the vehicle network based on the connectivity information includes configuring a firewall (e.g., on a

gateway node, a DC, or another node) in the vehicle network to secure intra-domain communications, inter-domain communications, or both)***; and***

wherein the gatekeeper interface circuit is structured to regulate communications between end points of the at least one network zone and the at least one of the external communication portal or the first external device {BARRETT, paragraph 0034, 0037, 0046, 0064 The firewall system can include one or more firewalls that can filter inbound and outbound network traffic based on predetermined rules, and disallow data transfers from unauthorized sources. The firewalls may perform context-aware filtering or any other types of advanced traffic filtering. For example, a firewall can be configured to allow packets of a type that appear on a whitelist to pass through the firewall, while blocking packets of a type that do not appear on the whitelist. In some implementations, a firewall can be configured to perform intrusion detection to detect when a packet is blocked by the firewall and then to log, report, and/or take other action). BARRETT does not explicitly disclose ***wherein the policy is communicated to the CND from a second external device***. However, UNAGAMI discloses ***wherein the policy is communicated to the CND from a second external device*** (UNAGAMI, paragraph 0052, wherein " In addition, the in-vehicle network system may further include an external communication electronic controller, and the updated rule information may be received by the gateway device in response to a receipt of the updated rule information by the external communication electronic controller from an external device, the updated rule information being transmitted on the first bus by the external communication electronic controller. Thus, rule information for detecting a malicious frame can be appropriately updated even if a device in an in-vehicle network system that communicates with an external device and a fraud-detection ECU have separate configurations").

Therefore, it would have been obvious to one ordinary skill in the art before the effective filing date of the invention to modify BARRETT using UNAGAMI so that the policy is revived from an external device as claimed because this would have provided a way to update the in-vehicle gateway/firewall with new rules that can help handle new security issues which improves the security of the system.

As pre claim 2, claim 1 is incorporated and UNAGAMI discloses **wherein the second external device comprises one of a cloud application or a web-based tool** (UNAGAMI, Fig 28, paragraph 0174, wherein "The in-vehicle network system 12 is configured to include fraud-detection ECUs 3100a to 3100c, a head unit 3200, gateways 3300 and 3300a, and nodes connected to buses, such as ECUs 400a to 400e connected to various devices. Of the constituent elements of the in-vehicle network system 12, constituent elements having functions similar to those illustrated in the first embodiment or the second embodiment are designated by the same numerals and are not described. The head unit 3200 is obtained by partially modifying the head unit 200 illustrated in the first embodiment, and is capable of, as illustrated in FIG. 28, communicating with an external device such as a server 600 via an external network 550");

As pre claim 8, claim 2 is incorporated and BARRETT discloses **wherein the policy comprises at least one data collection definition, each data collection definition comprising at least one local communicating device corresponding to at least one data collection parameter; wherein the policy formulation engine determines at least one of a source or destination definition for at least one of the at least one data collection definitions, wherein each of the at least one of a source or destination definitions comprises at least one of: a source identifier; a destination identifier; a source address; a destination address; a source port; or a destination**

port; and wherein the CND is further structured to perform a data collection operation in response to the data collection definition (BARRETT, paragraph 0034, 0037, 0046, 0064 The firewall system can include one or more firewalls that can filter inbound and outbound network traffic based on predetermined rules, and disallow data transfers from unauthorized sources. The firewalls may perform context-aware filtering or any other types of advanced traffic filtering. For example, a firewall can be configured to allow packets of a type that appear on a whitelist to pass through the firewall, while blocking packets of a type that do not appear on the whitelist. In some implementations, a firewall can be configured to perform intrusion detection to detect when a packet is blocked by the firewall and then to log, report, and/or take other action);

As pre claim 9, claim 8 is incorporated and BARRETT discloses **wherein the vehicle communication circuit is positioned on the first external device, wherein the CND is configured to provide the vehicle communications data to a port on a network zone of the vehicle, and wherein the first external device is configured to receive the communications data at the port on the network zone of the vehicle** (BARRETT, paragraph 0025, 0060, wherein "As illustrated in FIG. 1, the internal network 100 includes a connectivity domain 105, an in-vehicle experience domain 110, a driver replacement domain 120, a body and comfort domain 130, and a powertrain and vehicle dynamics domain 140. The connectivity domain 105 includes network nodes such as one or more of an antenna 115, a telematics unit 125, a central gateway 135, or an on-board diagnostics port 145. The connectivity domain 105 can provide communications between the internal network 100 of the vehicle and one or more external networks of the outside world, for example, through the antenna 115. The connectivity domain 105 can also provide communications between many functional domains and network nodes

(e.g., ECUs) of the internal network 150, for example, through an Ethernet switch 131 connected to the central gateway 135”};

As pre claim 10, claim 8 is incorporated and BARRETT discloses **wherein the visualization circuit is positioned on the first external device, and wherein the display interface circuit transmits the visualization data to a user device communicatively coupled to the first external device** (BARRETT, paragraph 0117, wherein “o provide for interaction with a user, implementations of the subject matter described in this specification can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube), LCD (liquid crystal display), LED (Light Emitting Diode), or plasma monitor, for displaying information to the user and a keyboard and a pointing device, e.g., a mouse, trackball, or trackpad by which the user can provide input to the computer. Input may also be provided to the computer using a touchscreen, such as a tablet computer surface with pressure sensitivity, a multi-touch screen using capacitive or electric sensing, or other type of touchscreen. Other kinds of devices can be used to provide for interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser”};

As pre claim 11, claim 8 is incorporated and BARRETT discloses **wherein the visualization data comprises at least one of: a graphical flow representation of at least a**

portion of communications between local communicating devices of the vehicle; a graphical flow representation of at least a portion of communications passing through the CND; a graphical flow representation of at least a portion of communications regulated by at least one of the first network interface circuit or the second network interface circuit; a graphical flow representation of at least a portion of communications passing between the first network zone and the second network zone; and a graphical flow representation of at least a portion of communications with the at least one of the external communications portal or the first external device (BARRETT, paragraph 0117, wherein “o provide for interaction with a user, implementations of the subject matter described in this specification can be implemented on a computer having a display device, e.g., a CRT (cathode ray tube), LCD (liquid crystal display), LED (Light Emitting Diode), or plasma monitor, for displaying information to the user and a keyboard and a pointing device, e.g., a mouse, trackball, or trackpad by which the user can provide input to the computer. Input may also be provided to the computer using a touchscreen, such as a tablet computer surface with pressure sensitivity, a multi-touch screen using capacitive or electric sensing, or other type of touchscreen. Other kinds of devices can be used to provide for interaction with a user as well; for example, feedback provided to the user can be any form of sensory feedback, e.g., visual feedback, auditory feedback, or tactile feedback; and input from the user can be received in any form, including acoustic, speech, or tactile input. In addition, a computer can interact with a user by sending documents to and receiving documents from a device that is used by the user; for example, by sending web pages to a web browser on a user's client device in response to requests received from the web browser”);

As pre claim 12, claim 1 is incorporated and BARRETT discloses wherein a first network zone of the at least one network zone comprises a controller area network (CAN) and wherein

a second network zone of the at least one network zone comprises an ethernet network zone (BARRETT, paragraph 0040, wherein "The powertrain and vehicle dynamics DC 450 can support communications based on different communication protocols such as TCP/IP stack 422, CAN stack 432, and FlexRay stack 443. For example, the powertrain and vehicle dynamics DC 450 can be connected to an Ethernet that includes the Ethernet switch 445 for inter-domain communication with other domains 415, 425, and 435. The powertrain and vehicle dynamics DC 450 can be connected with a CAN network that includes one or more actuator ECUs 432, 436 and sensor ECUs 434, 438 that support CAN communications. In some implementations, CAN is used, for example, for medium-speed (e.g., 1 Mbps) applications including ECU-to-ECU communications");

As pre claim 13, claim 2 is incorporated and UNAGAMI discloses **wherein the second external device comprises a policy formulation engine structured to generate the policy as at least one configuration file, the at least one configuration file having a selected configuration format** (UNAGAMI, paragraph 0052, wherein " In addition, the in-vehicle network system may further include an external communication electronic controller, and the updated rule information may be received by the gateway device in response to a receipt of the updated rule information by the external communication electronic controller from an external device, the updated rule information being transmitted on the first bus by the external communication electronic controller. Thus, rule information for detecting a malicious frame can be appropriately updated even if a device in an in-vehicle network system that communicates with an external device and a fraud-detection ECU have separate configurations");

As pre claim 14, claim 2 is incorporated and UNAGAMI discloses **wherein the second external device comprises: a policy interaction engine structured to generate policy interaction code; a policy development interface structured to accept policy input values from a plurality of users in response to the policy interaction code; a policy formulation engine structured to generate the policy in response to accepted policy input values; and a policy application engine structured to communicate the policy to the CND** (UNAGAMI, paragraph 0052, wherein " In addition, the in-vehicle network system may further include an external communication electronic controller, and the updated rule information may be received by the gateway device in response to a receipt of the updated rule information by the external communication electronic controller from an external device, the updated rule information being transmitted on the first bus by the external communication electronic controller. Thus, rule information for detecting a malicious frame can be appropriately updated even if a device in an in-vehicle network system that communicates with an external device and a fraud-detection ECU have separate configurations");

As pre claim 15, claim 14 is incorporated and UNAGAMI discloses **wherein the policy interaction code comprises at least one of: a header file; a parameter definition; or an application programming interface (API) declaration** (UNAGAMI, paragraph 0052, wherein " In addition, the in-vehicle network system may further include an external communication electronic controller, and the updated rule information may be received by the gateway device in response to a receipt of the updated rule information by the external communication electronic controller from an external device, the updated rule information being transmitted on the first bus by the external communication electronic controller. Thus, rule information for detecting a malicious frame can be appropriately updated even if a device in an in-vehicle

network system that communicates with an external device and a fraud-detection ECU have separate configurations"};

Conclusion

Any inquiry concerning this communication or earlier communications from the examiner should be directed to HAMZA N ALGIBHAH whose telephone number is (571)270-7212. The examiner can normally be reached on 7:30 am - 3:30 pm.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Umar Cheema can be reached on (571) 270-3037. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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